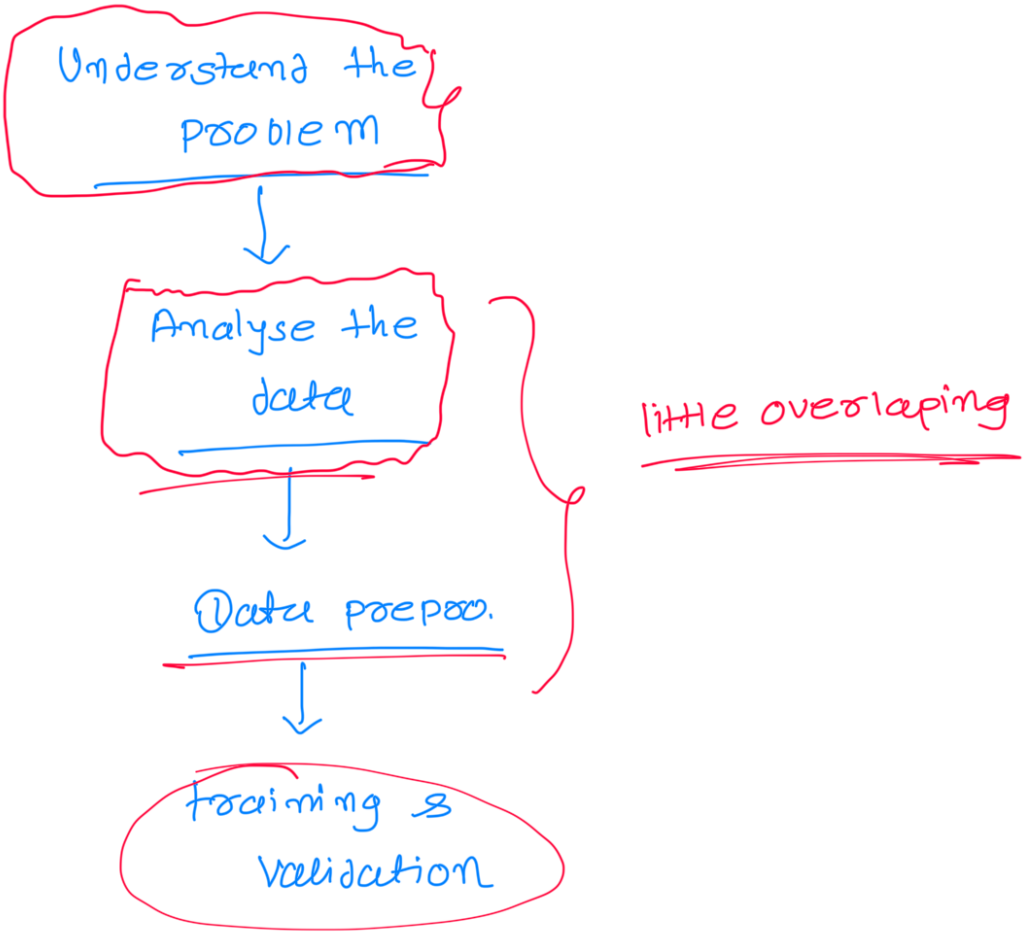
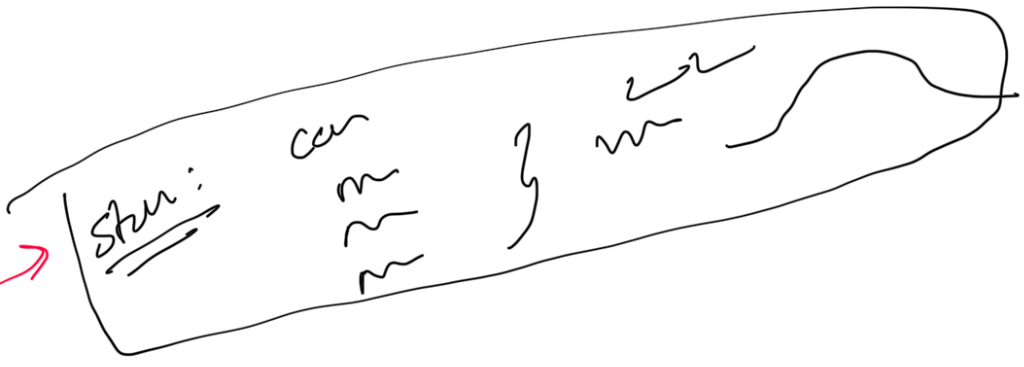
* ML workflow:



* Small topics Just go through that:

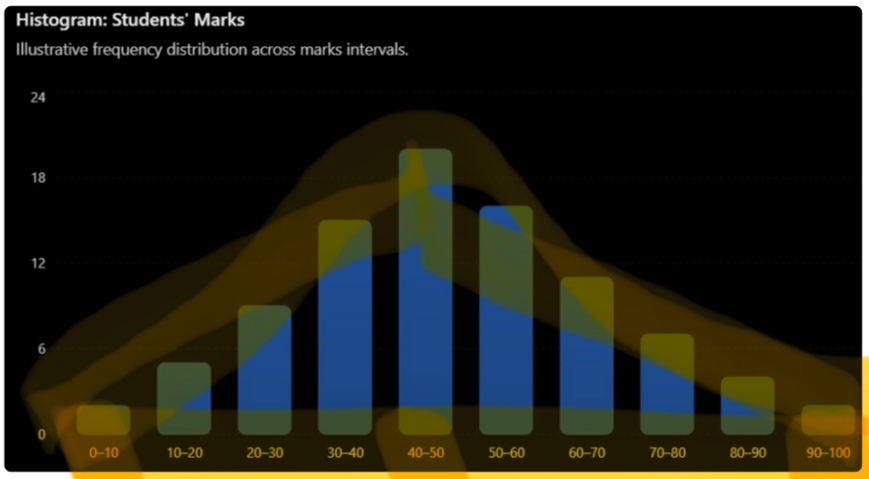
- Json
- CSV data
- SQL
- Tensor

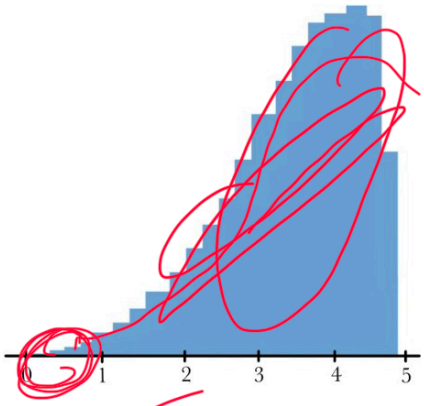


* EDA:
  - ↳ Plots

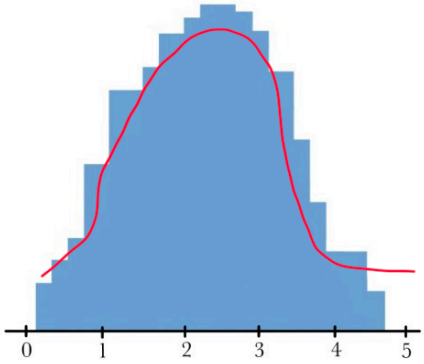
(1) Histogram: How Numerical data is distributed across

ranges, we can see shape, center & spread.

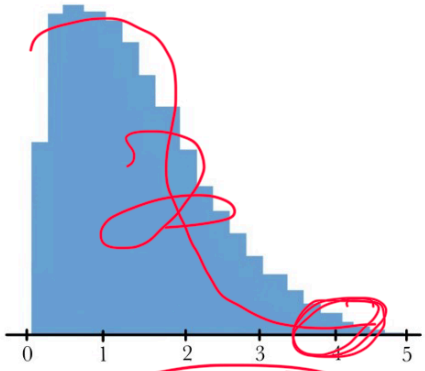




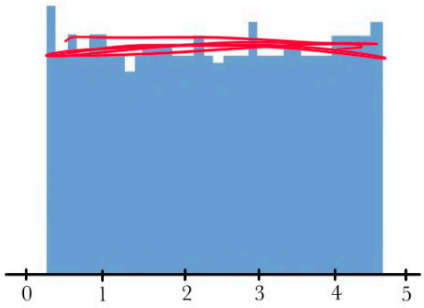
skew left



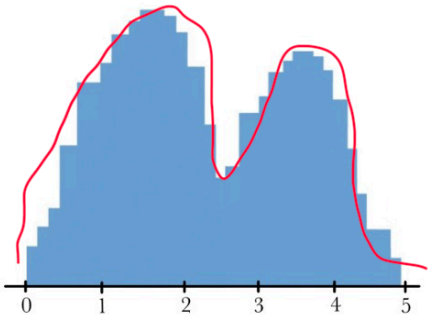
symmetric, unimodal



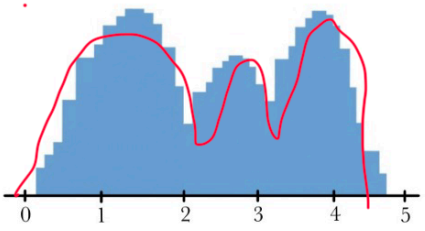
skew right



uniform



bimodal



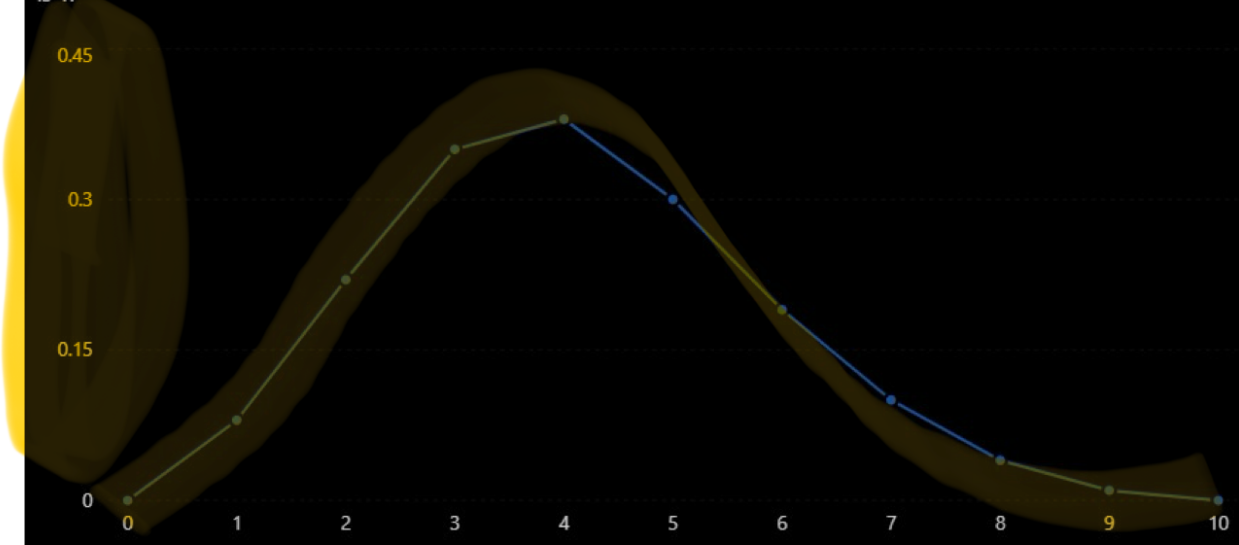
multimodal

② Desity plot : (Pro. distribution)

- Distribution
- Skewness
- Peaks
- Compare groups.
- Symmetric

Density Plot

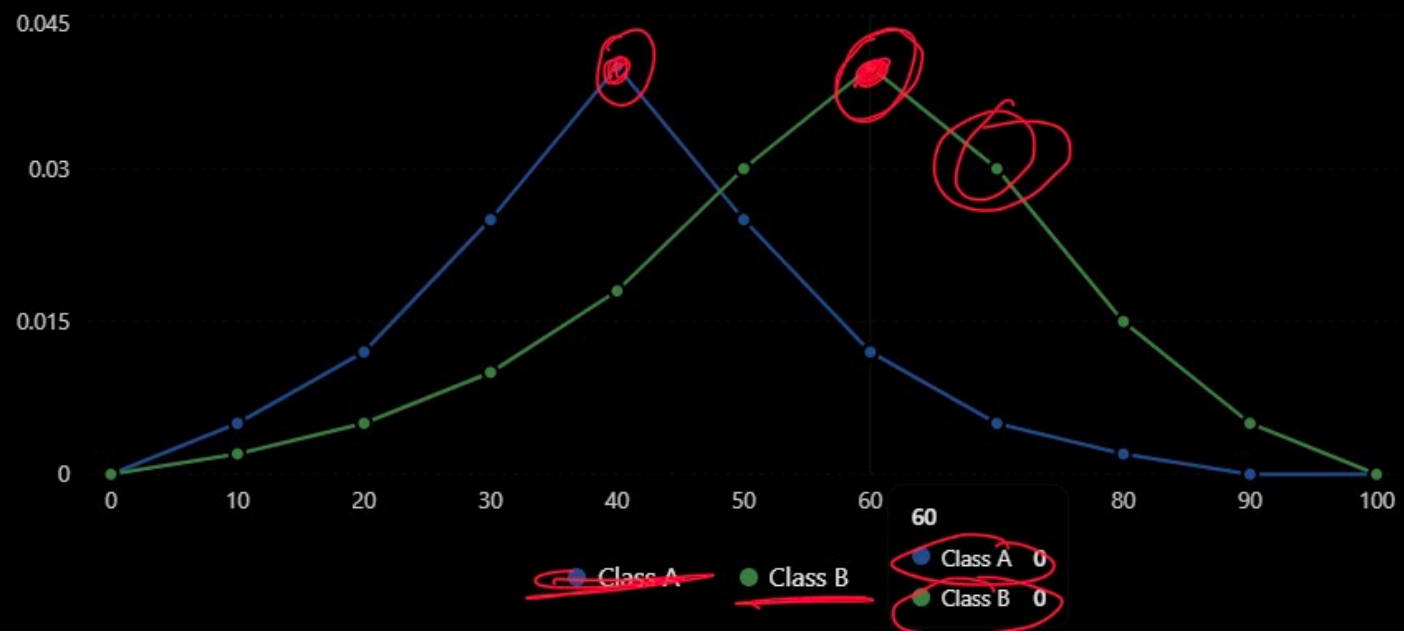
Illustrative probability density curve for numerical data. The total area under a valid probability density curve is 1.



General use: A density plot helps you understand where data values are concentrated, the shape of the distribution, and whether it is symmetric or skewed.

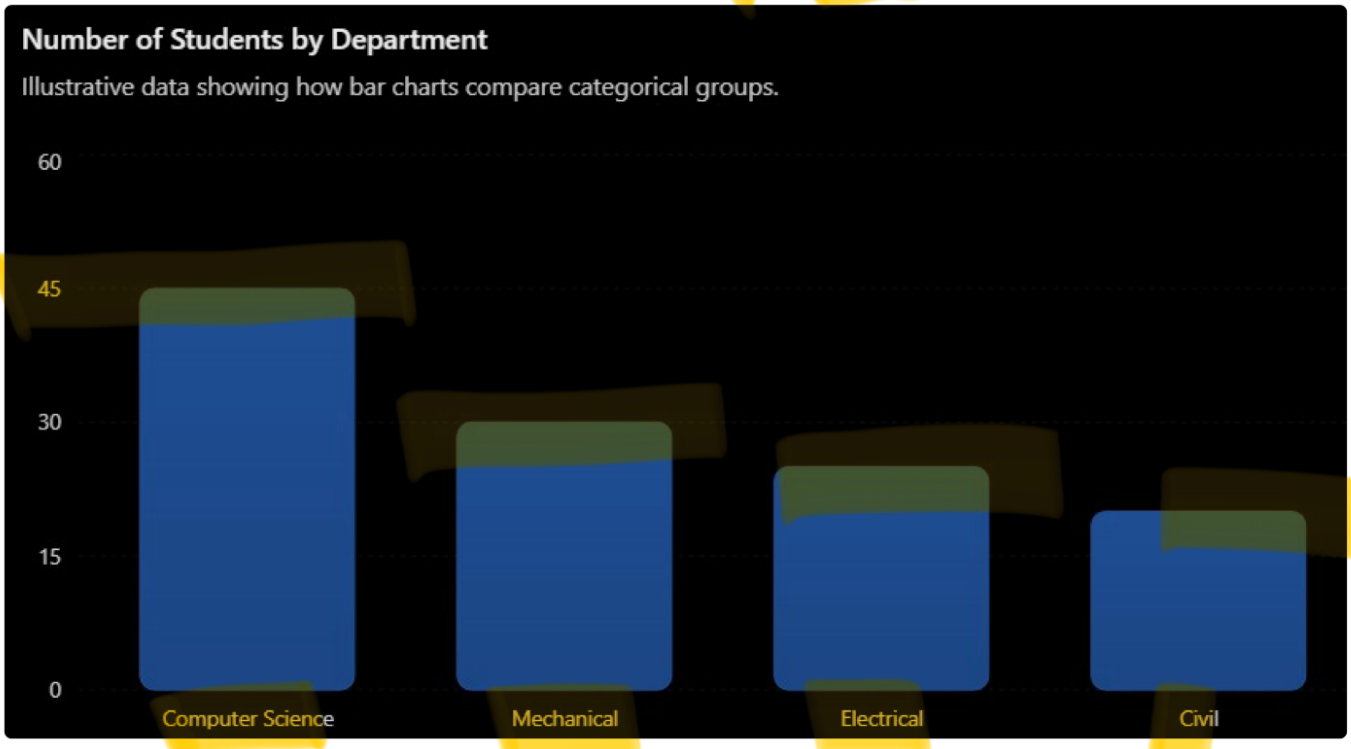
Comparing Exam Marks: Class A vs Class B

Illustrative density curves comparing the distribution of marks in two classes. Higher peaks indicate more concentrated values; curve position shows where marks tend to fall.



③ Bar chart:

- compare values across categories.

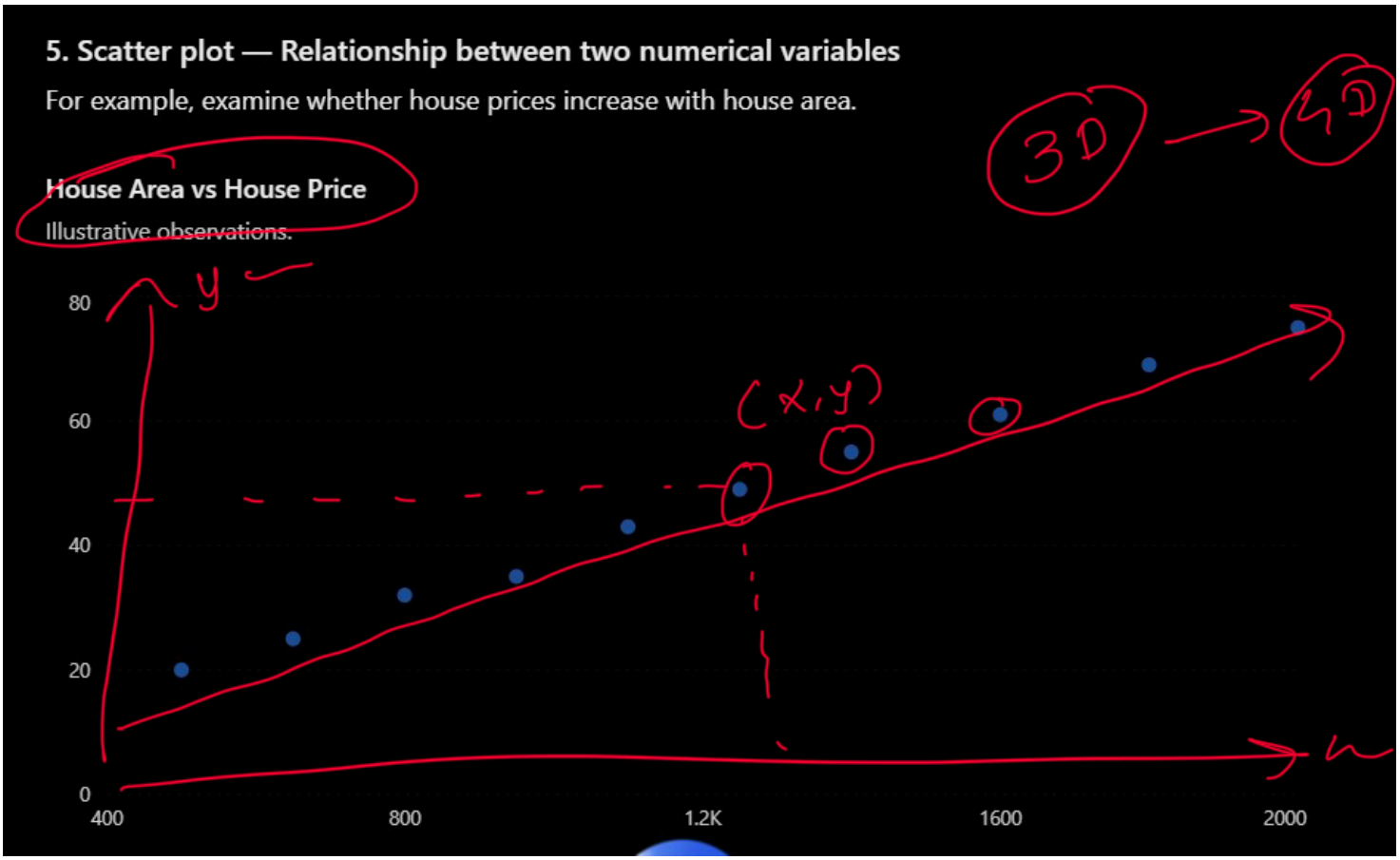


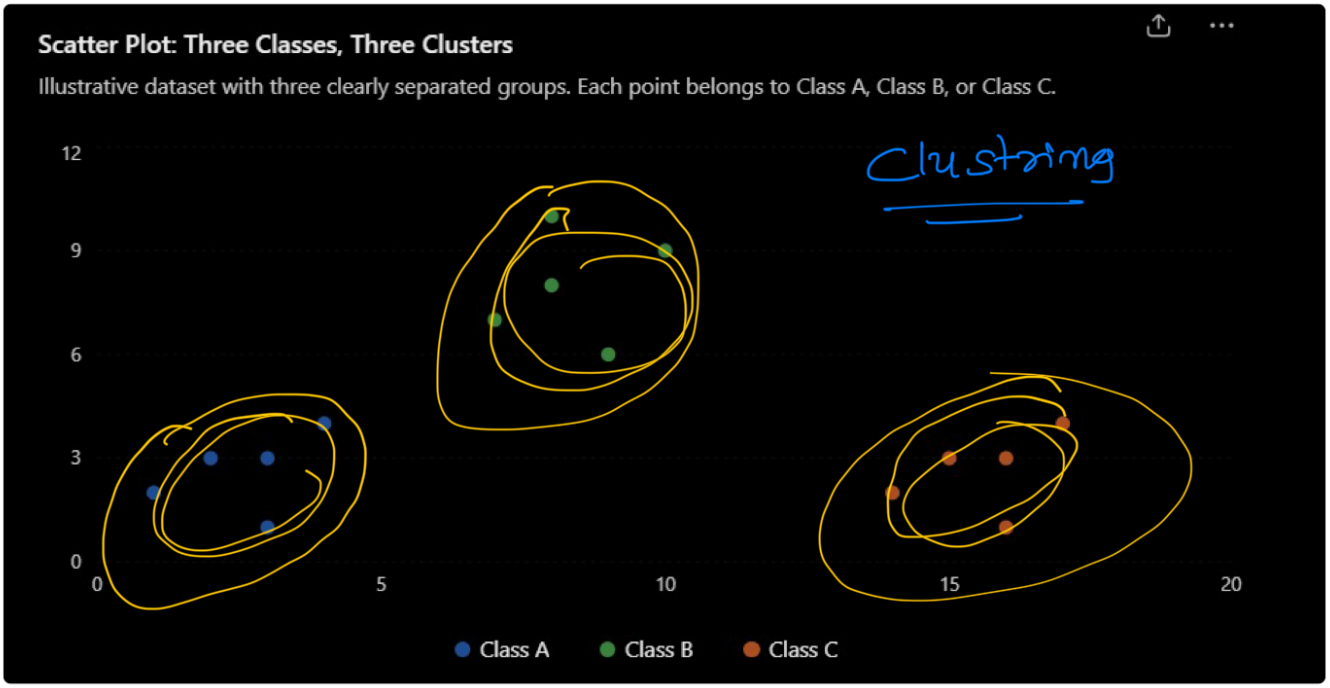
④ Scatter plot:

- Relationship btwn two numerical variables (correlation visualisation)

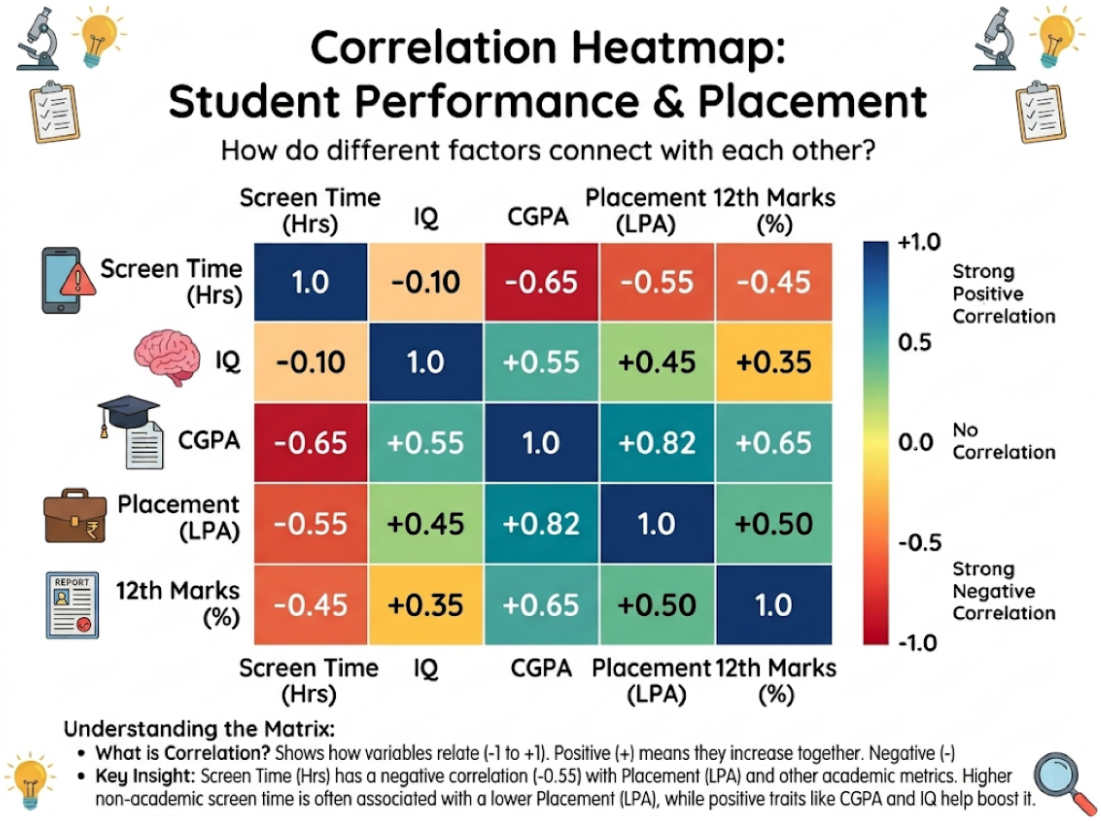
5. Scatter plot — Relationship between two numerical variables

For example, examine whether house prices increase with house area.





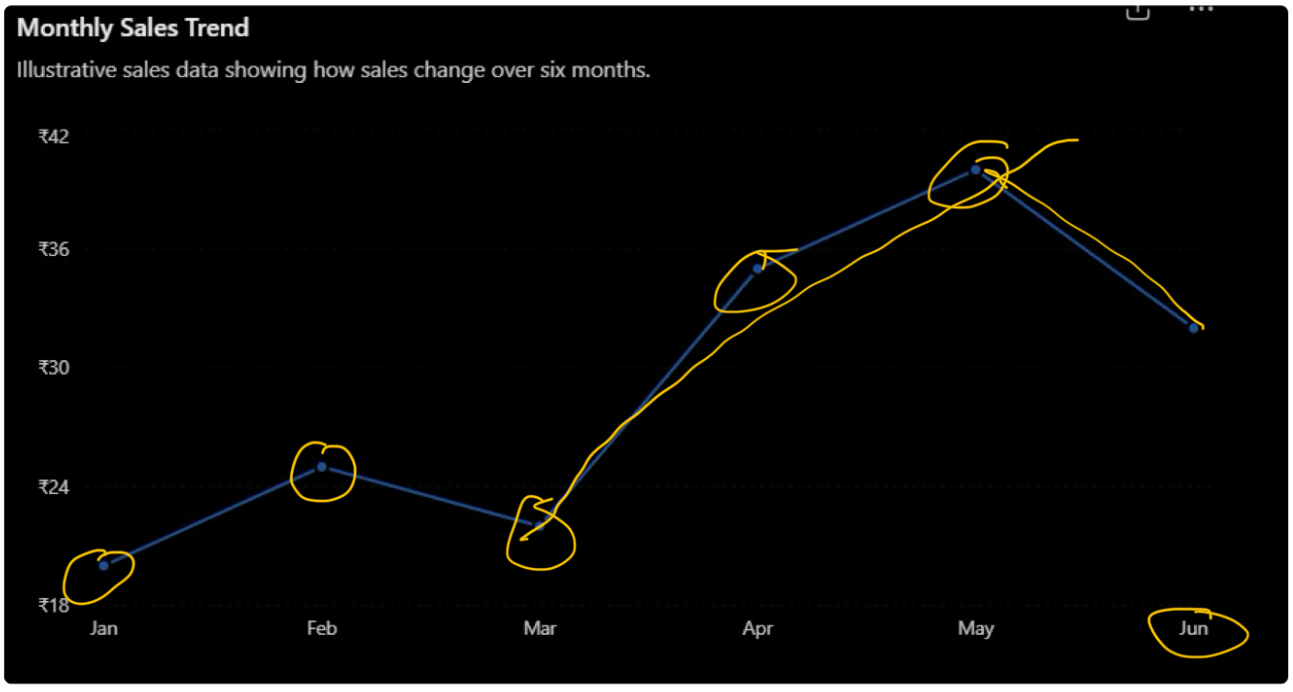
5 Heatmap: corelation analysis

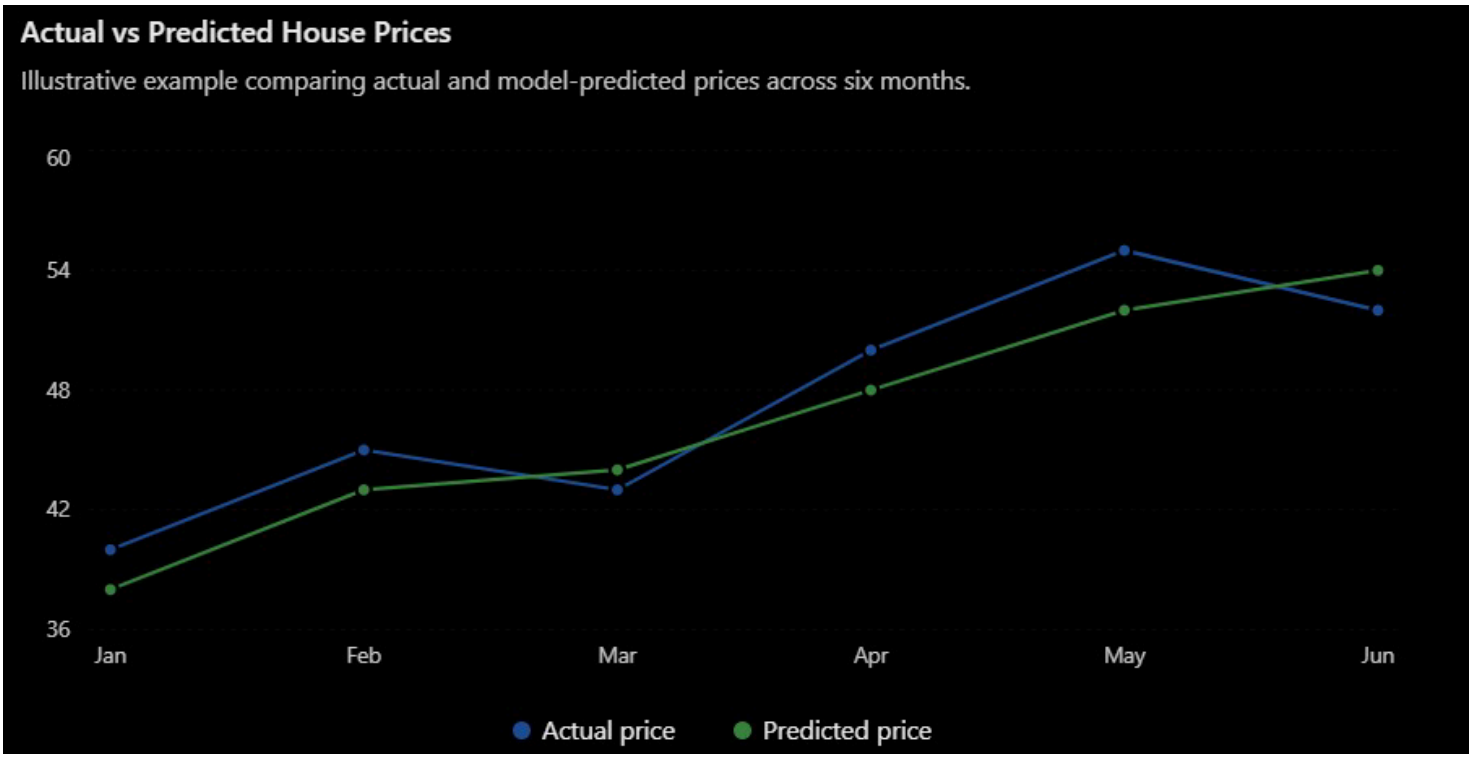


(6) Line plot:

How something changes over time.

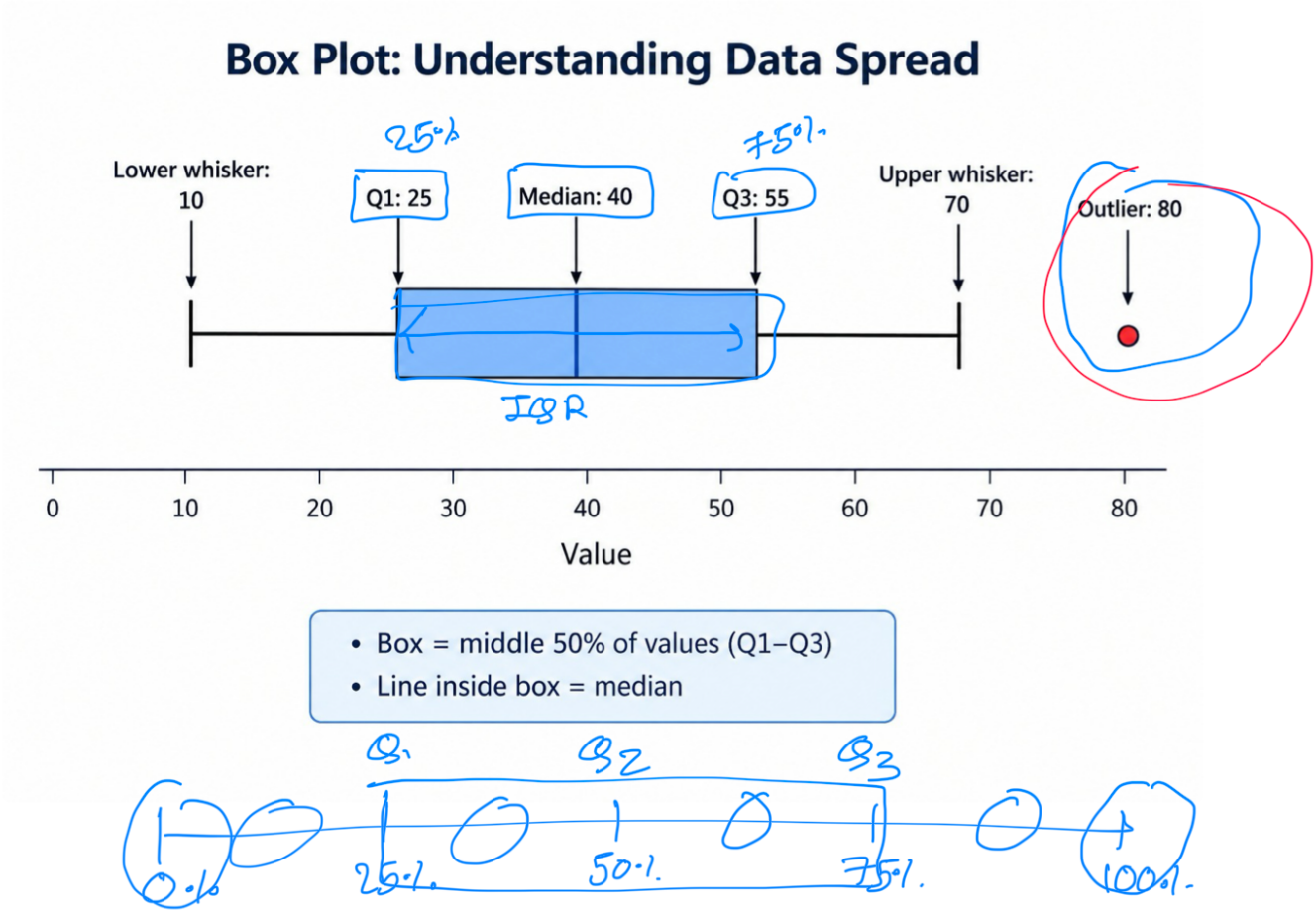
(trend analysis)




Actual vs Predicted House Prices
Illustrative example comparing actual and model-predicted prices across six months.
60
54
48
42
36
Jan
Feb
Mar
Apr
May
Jun
Actual price
Predicted price

⑦ Box plot: → [Graph (1

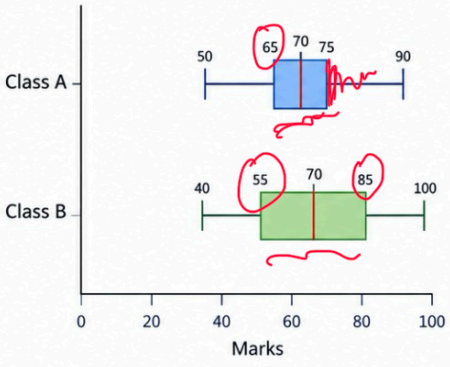
- median
- spread
- {Outliers}



Box Plot: 3 Examples for Better Understanding

A box plot helps us see the median, spread and possible outliers in data.

1. Exam marks: compare classes



Same median, different spread

Both classes have a similar median (around 70), but Class B has a wider spread, meaning more variation in marks.

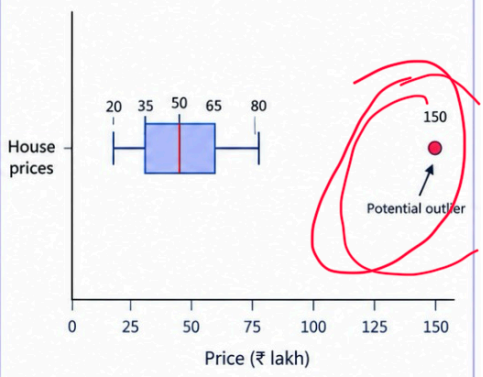
2. Salaries: compare departments



Different typical salaries and variability

Sales has a higher median salary and a wider spread, while Support has a lower median and a narrower spread.

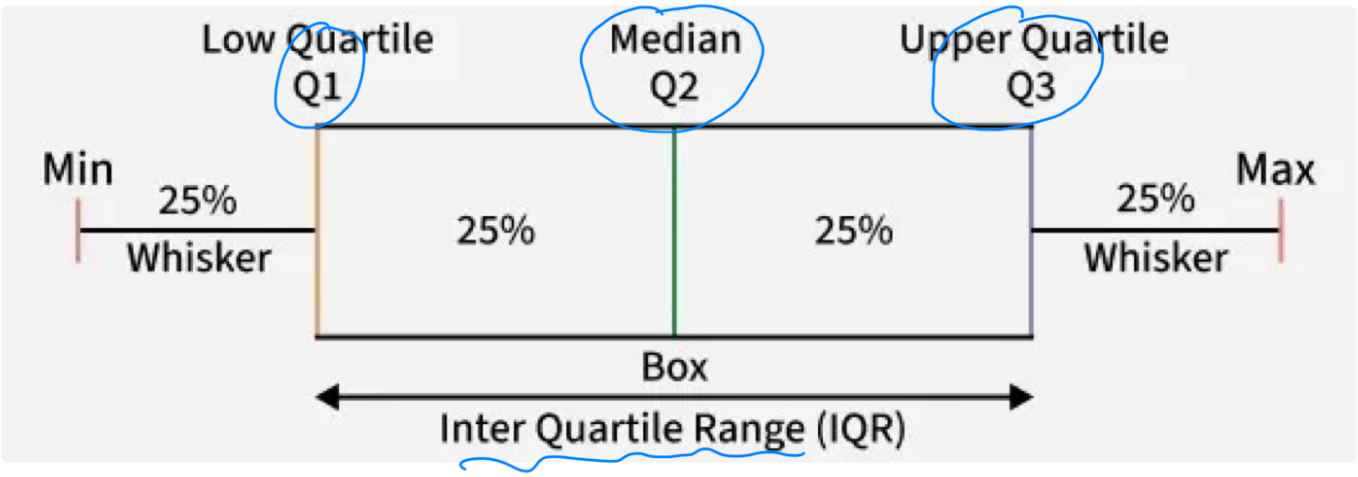
3. House prices: spot an outlier



A very high price stands apart

Most houses are between 20 and 80 lakh, with a median of 50 lakh, but one house is priced much higher (150 lakh).

* How to draw a Box plot:
(and How to detect outliers using that)



Q_1 : 25 % tile

Q_2 : 50 % tile

Q_3 : 75 % tile

$IQR = Q_3 - Q_1$

$1.5 * IQR + Q_3$ → 100% tile ?

$Q_1 - 1.5 * IQR$

middle (median) 50% tile

1 2 3 4 ... 25 ... 50 .... 75 .... 98 99 100 101

25% tile

75% tile

98%

76%

Value of percentile (location) $= \frac{P(N+1)}{100}$

N = total no.

75% tile $= \frac{75(101)}{100} = 75.75$

→ IQR (Inter quartile Range) $= Q_3 - Q_1$

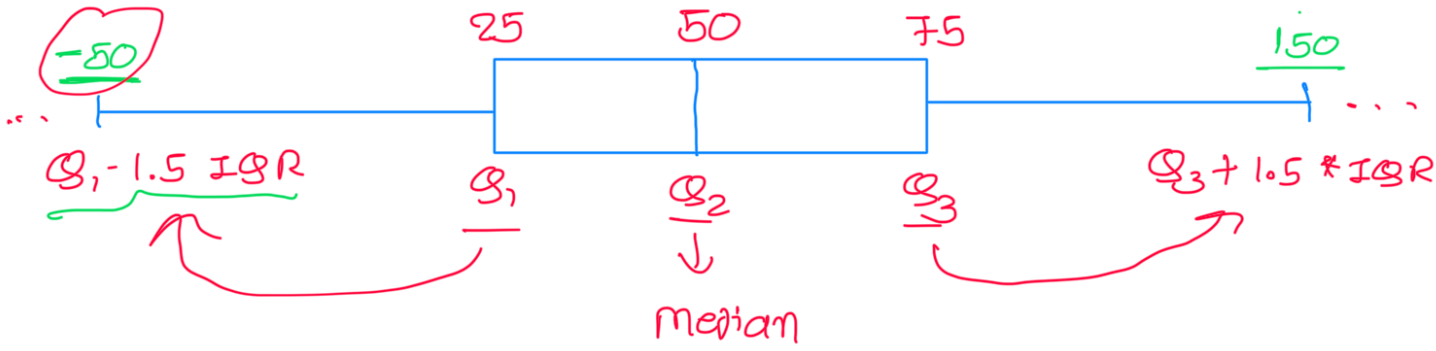
In our Example = 75 - 25 = 50

→ lower bound = $Q_1 - 1.5 * IQR$

→ Upper bound = $Q_3 + 1.5 * IQR$

$25 - 1.5 * 50$

$75 + 1.5 * 50$



→ Outliers: $x > Q_3 + 1.5 * IQR$

$x < Q_1 - 1.5 * IQR$

} is it strickt Rule ?

⇒ How to detect outliers: IQR method

→ is there any other methods for same ?

- Scatter plot
- Z score method

* is outlier is good or Not ?

* How to handle outliers:

① **Remove / Trimming:**

- when outliers are Errors / irrelevant

② **Caping:**

- Replace extreme values with choosen lower & upper limits.

Ex: if x = 180 (upper limit 150)

↳ x = 150

③ **Imputation:** → put down

- Replace value with median or another Imputer.

④ **Transformation:** later

⑤ **Keep the outliers**

⑤ Verify it with domain Expert.

⑥ Other methods!

MAE how

MSE ←

- Use models or loss functions less sensitive to extreme values.
- Special treatment like grouping or some.
- Add an outlier indicator.

2nd Missing Value Analysis:

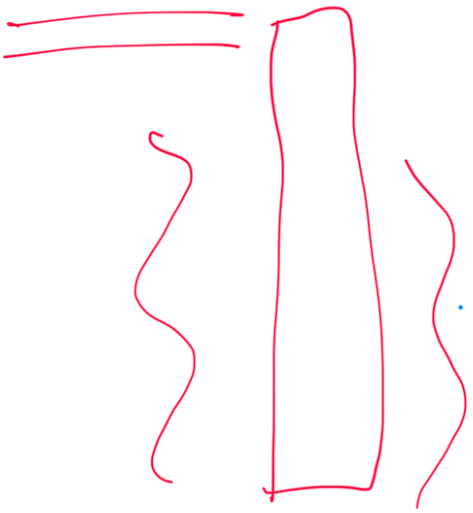
- Some ML algo can not handle missing.
- Wrong imputation can distort data.

→ How to handle missing value.

① Row deletion ✓

② column " ✓

③ mean Imputation



(4) mode ''

(5) median ''

(6) constant value ''

(7) group wise imputation: Fill mis. val. using some imputation techn. within a group.

Ex: Use the median salary of same Job role.

(8) Random

(9) KNN, Regression, classification ... Imputation.

(10) Missing Indicator: Add new binary column

(11) Some algo handle m.v. Internally:
- xg boost
- cat boost
- Light GBM

(12) Treat missing as category → (3)

(13) Domain based